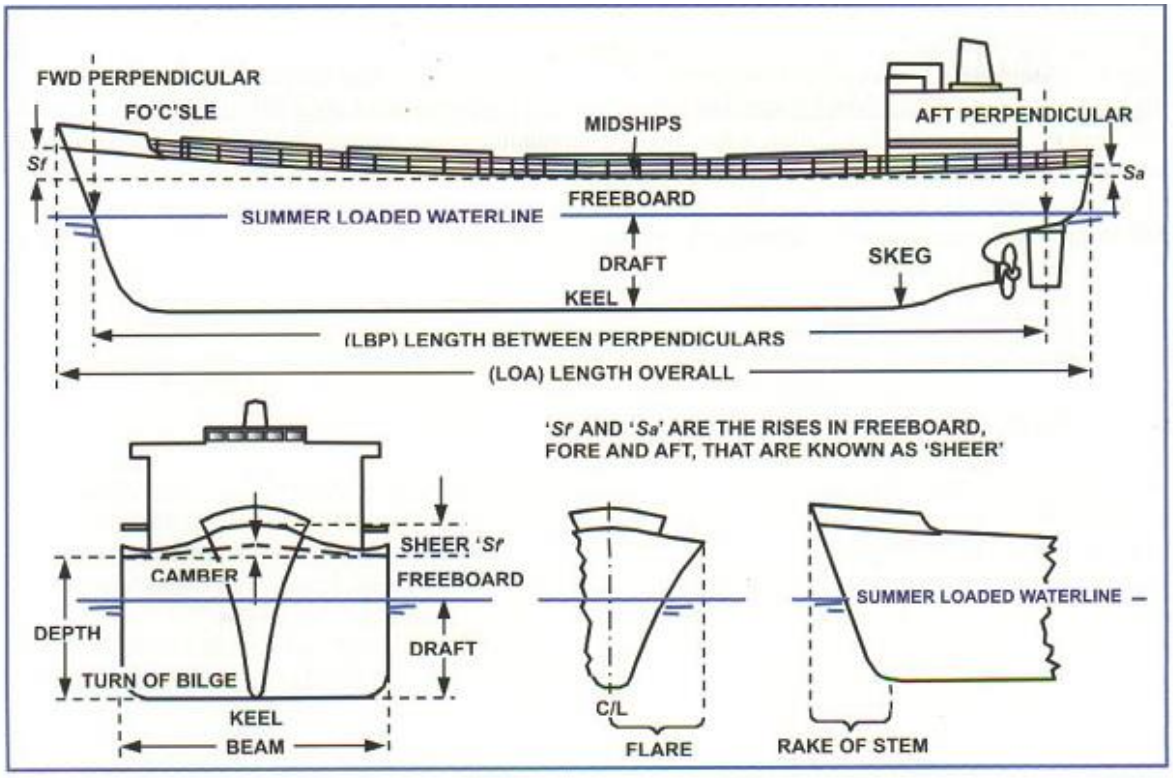
BASIC OF NAVAL ARCHITECTURE



Fore perpendicular: It is the vertical line passing through the point of intersection of the summer load line with forward side of the stem.

Aft perpendicular: It is the line perpendicular to the waterplane and passing through the centerline of the rudder pintles.

Length between perpendiculars (lbp): The distance measured along the summer load water plane from fore to aft perpendicular is known as *length between perpendiculars*.

Length overall (loa): It is the distance measured parallel to the summer load waterline between the extreme points at the forward and the aft. The extreme point at the forward can be taken on the bulbous bow.

Length on the waterline (lwl): It is the distance measured between the intersections points of the bow and the after end with the summer load waterline. In case it is not mentioned, the length of the summer load waterline is taken into consideration as the length of the waterline.

Midship or amid ship : The midpoint between the aft and the forward perpendicular is known as midship or amid ship. The section passing through this point and which is normal to the waterplane is called *midship section.* The term 'beam' is quoted at amid ship.

The *breath extreme* is measured at amid ship. It is also known as *moulded breadth line.*

Depth: The depth of the ship varies along the length. But the depth of the ship is taken as the distance between the undersides of the deck amid ship to the bottom of the keel. You might not get an exact value of depth as the hull is symmetrical and the depth varies thorough out the ship. For example, the depth is greater at stern than at midship.

Width or beam or extreme breadth: The width of the ship is also known as the *extreme breadth*. It is generally measured at the amid ship. It is the greatest distance between the two sides of the ship at the greatest width.

Sheer: It is measured as the rise of the deck towards the stem or stern. It is the height of the deck at the side above the deck at sides amidships.

Camber or round of beam: It can be defined as the side of the deck going from side to the centre of the ship.

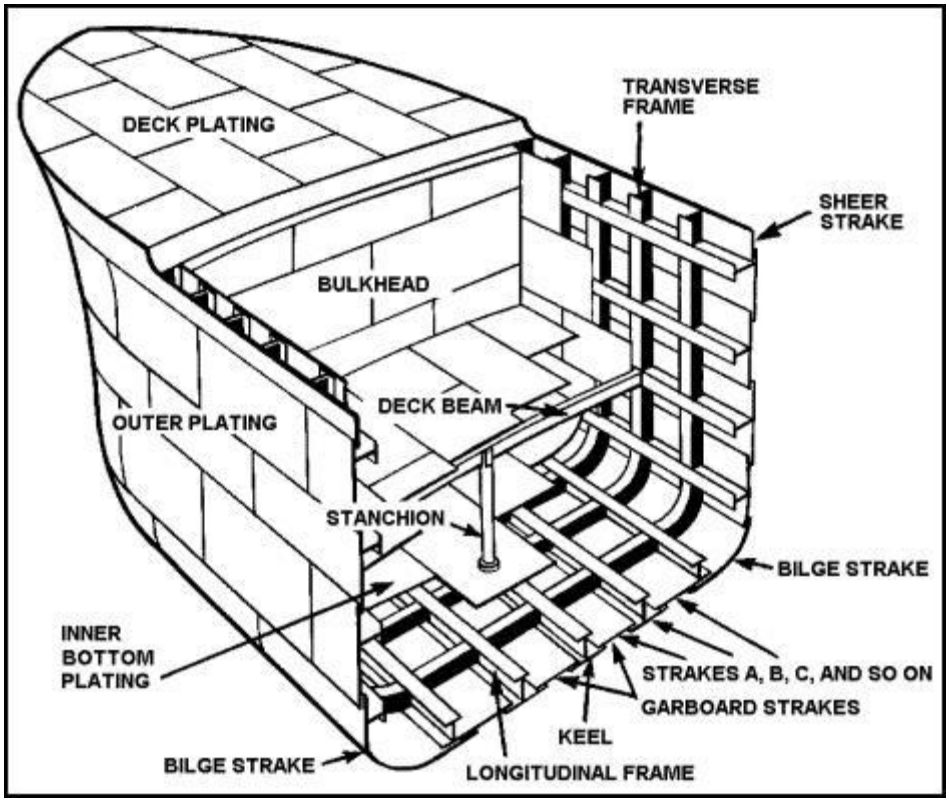
Rise of floor or dead rise: At the amidships region, the bottom of the ship is extended out to intersect the *moulded breadth line*. The rise or the height of this intersection above the keel is known as *rise of floor or dead rise*.

Tumble home: If the sides of the ship at the amidships is not vertical and if the upper deck beam is less than the waterline, it is said to have ***tumble home***.

Draught: It is the distance between the keel and the waterline at any point along the length of the ship. *Moulded draught* is measured from the inside of the keel plating.

Trim: The difference between the draughts at forward and aft is known as the *trim*. Trim is mentioned as by the bow or by the stern depending upon which one is greater.

Freeboard: It is essential in determining the stability of the ship. It is the difference between the depth at side and the draught, i.e. it is the height of the deck above the waterline. Freeboard is generally less at amidships than at bow and stern.



COEFFICIENTS OF FORM - Coefficients of form are dimensionless numbers that describe hull fineness and overall shape characteristics. The coefficients are ratios of areas or volumes for the actual hull form compared to prisms or rectangles defined by the ship's length, breadth, and draft. - Since length and breadth on the waterline as well as draft vary with displacement, coefficients of form also vary with displacement. - Tabulated coefficients are usually based on the molded breadth and draft at designed displacement. Length between perpendiculars (Lbp) is most often used, although some designers prefer length on the waterline. - Coefficients of form can be used to simplify area and volume calculations for stability or strength analyses. - As hull form approaches that of a rectangular barge, the coefficients approach their maximum value of 1.0.

BLOCK COEFFICIENT (Cb) block coefficient (Cb) is the ratio of the immersed hull volume (∇) at a particular draft to that of a rectangular prism of the same length, breadth, and draft as the ship: $Cb = \nabla/LBd$

where:

∇ == immersed volume,

B = beam,

d = draft L = length between perpendiculars

MIDSHIP SECTION COEFFICIENT (Cm) The midship section coefficient (Cm) is the ratio of the area of the immersed midship section (Am) at a particular draft to that of a rectangle of the same draft and breadth as the ship: Cm= Am / B x d

Where,

Am = area of the immersed portion of the midships section, [m2]

B = beam, usually taken at the waterline, [m]

d = draft, [m]

WATERPLANE AREA COEFFICIENT (Cw) The waterplane area coefficient (Cw) is the waterplane area divided by the length x the breadth. So, Cw= Aw/L x B

Where,

Cw= Waterplane Area (m2)

L= Length (m)

B= Breadth (m)

THE PRISMATIC COEFFICIENT (Cp) The prismatic coefficient (Cp) is the underwater area, divided by the area of a midship section, times the length of the ship. It is an indication of hull fineness, and may be broken down into fore & aft components. This formula can be written as Cp = ∇/Am x L

where, = immersed volume, [m3]

Am = area of the immersed portion of the midships section, [m2]

L = length between perpendiculars,

We Know

, ∇= Cb x L x B x d

and Am= Cm x B x d

Now, Cp = ∇/Am.L So,

Cp= Cb. L.B.d/Cm.B.d.L

Cp= Cb/Cm

Some typical values are presented in the table below:

Type of vessel	*Block coefficient*	*Prismatic coefficient*	*Midship area coefficient*
Crude oil carrier	0.82–0.86	0.82–0.90	0.98–0.99
Product carrier	0.78–0.83	0.80–0.85	0.96–0.98
Dry bulk carrier	0.75–0.84	0.76–0.85	0.97–0.98
Cargo ship	0.60–0.75	0.61–0.76	0.97–0.98
Passenger ship	0.58–0.62	0.60–0.67	0.90–0.95
Container ship	0.60–0.64	0.60–0.68	0.97–0.98
Ferries	0.55–0.60	0.62–0.68	0.90–0.95
Frigate	0.45–0.48	0.60–0.64	0.75–0.78
Tug	0.54–0.58	0.62–0.64	0.90–0.92
Yacht	0.15–0.20	0.50–0.54	0.30–0.35